

Flowed Metal: Micro-particle Impacts Metallic Glasses under High Temperature

Abstract

Bulk metallic glasses (BMGs) represent a unique class of materials that combine the disordered atomic structure of glass with the properties of metals. Their response to dynamic loading, such as high-velocity impact, is not only crucial for potential structural applications but also fundamental to understanding non-crystalline physics. Crystalline metals show a counterintuitive phonon-drag strengthening effect under high strain rates at high temperatures, which contrasts with glasses conventional thermal softening behavior. Therefore, whether BMGs exhibit similar phenomenon remains an open question. In this work, the impact response of a $\text{Pd}_{43}\text{Ni}_{10}\text{Cu}_{27}\text{P}_{20}$ BMG at 25 °C, 100 °C and 200 °C was investigated via laser-induced particle impact tests and finite element method simulations. In-situ observation revealed a significant enhancement in energy dissipation of BMGs at 200 °C. Post-impact characterization identified a transition in deformation mechanism: solid-like craters with minimal pile-up at 25 °C gave way to fluid-like, irregular pile-up morphologies at 200 °C, suggesting adiabatic heating-induced softening. A calculated $\sim 10.4\%$ reduction in dynamic yield strength at high temperature provided quantitative confirmation of this thermal softening. Furthermore, simulations incorporating a physical constitutive model successfully replicated the experimental results and uncovered the underlying mechanism: high temperature promotes the activation of shear transformations, leading to greater plastic flow and prolonged impact interaction time. This study conclusively demonstrates that temperature is a critical, governing factor for the dynamic performance of BMGs, challenging simplistic material selection and providing vital insights for their application in extreme environments.

Key Words: Micro-particle impact, Metallic glass, Temperature, Plastic flow, Energy dissipation

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1. Background and Motivation

Originating in antiquity from molten sand, glass has been reshaped over centuries by technological advances into a crucial material across many fields. It is a material well known for its special properties, which are generated by its unique structure. Although glass is a solid at room temperature, it exhibits the characteristics of a super-viscous liquid at high temperature^[1]. This is because glass has an amorphous structure, in which the internal atomic arrangement is topologically random and disordered, as shown in Figure 1a. Due to the constraints of chemical bonds, the atomic configuration in glass shows a pattern of “short-range order and long-range disorder”. In contrast, another indispensable material for construction and modern technology, metal, is usually crystalline. Metals’ internal atomic arrangement is ordered, showing a high order of long-range periodicity, as shown in Figure 1b. Such a structure enables metals to have high strength and ductility. Although metals and glass seem to be two completely different materials, scientists have combined these two and produced a new material called metallic glass.

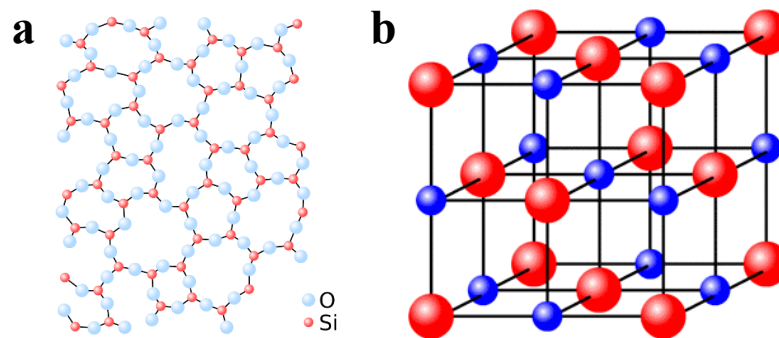


Figure 1 Topological structure of atomic packing: **a** molecular configuration of a silica glass^[20]

b atomic configuration of a GeTe alloy^[21]

Metallic glass (also called amorphous alloy) is produced using rapid solidification metallurgical technology. The “metal” refers to the fact that this material is composed of metallic elements; while the “glass” refers to the fact that this material has a glassy structure, meaning its atomic configuration is disordered, different from ordinary metals^[2]. When cooling molten metal extremely rapidly, its internal atoms do not have time to form an ordered crystalline structure, this prevents the formation of internal

crystalline phases and thus “freezes” the disordered state of the molten metal atoms. Consequently, this metal behaves more like a super-viscous liquid on microscopic scale^{[3][4]}.

The phenomenon of impact can be seen in every part of people’s daily life: water splash and deformation of car in accident are typical examples. Usually, glass underwent brittle damage under impact and shatters into pieces; metal, on the other hand, performs a better plastic deformation. Whereas amorphous alloy possesses the disorder structure of glass as well as performs the properties of metal. Thus, evaluating the impact response of this material is novel and interesting. Besides, as mentioned before, glass is softened and presents the characteristic of viscous liquid at high temperature. Recent research on metal shows that, under high strain rates ($>10^4 \text{ s}^{-1}$), the dynamic yield strength and dynamic hardness of pure metals (Cu, Au, and Ti) both increased with rising temperature due to the effect of phonon-drag strengthening^[5]. This differs from previous studies, where under moderate strain rates, increasing temperature typically softens metals^[6]. This shows that temperature has distinct but significant effects on metals under different strain rates. Meanwhile, studying the response of metallic glass to external energy inputs (such as temperature and mechanical force) plays a crucial role in developing non-crystalline physics and producing high-performance bulk metallic glass materials (BMGs). Inspired by these findings, this study investigates the mechanical response of BMGs under high-speed impact.

In this work, experiments of on high-velocity microparticle impacts on BMGs at different temperatures were conducted by laser-induced particle impact tests (LIPIT). The targets were examined to analyze the relationship between impact resistance ability and temperature. The finite element method (FEM) simulation was also carried out by using ABAQUS software to further reveal the underlying mechanisms.

2. Experiment

The study aims to investigate the deformation of BMGs under high-velocity microparticle impact and clarify the energy-dissipation mechanisms. Temperature is

treated as the independent variable; accordingly, two classical experimental groups were established: one at room temperature (approximately 25 °C) and the other at a high temperature of 200 °C. All other experimental parameters, including microparticle material and size, impact velocity and impact angle were held constant in each trial to ensure comparability and accuracy of the results.

As shown in Figure 2, the procedure was as follows: first, bulk samples of the $\text{Pd}_{43}\text{Ni}_{10}\text{Cu}_{27}\text{P}_{20}$ metallic glass were prepared through cutting and polishing. The samples were then examined by X-ray diffraction to confirm their amorphous structure. Differential Scanning Calorimetry (DSC) was also performed to determine the glass-transition temperature and the crystallization temperature of $\text{Pd}_{43}\text{Ni}_{10}\text{Cu}_{27}\text{P}_{20}$. Next, using a LIPIT system, the microparticles were launched to various velocities and impacted at the target. A high-speed camera (max 5,000,000 fps) recorded the impact process. After the impact, samples were examined by Scanning Electron Microscope (SEM) to reveal the target deformation; crater topography was then obtained via Laser Confocal Microscopy (LCM) to validate the simulation.

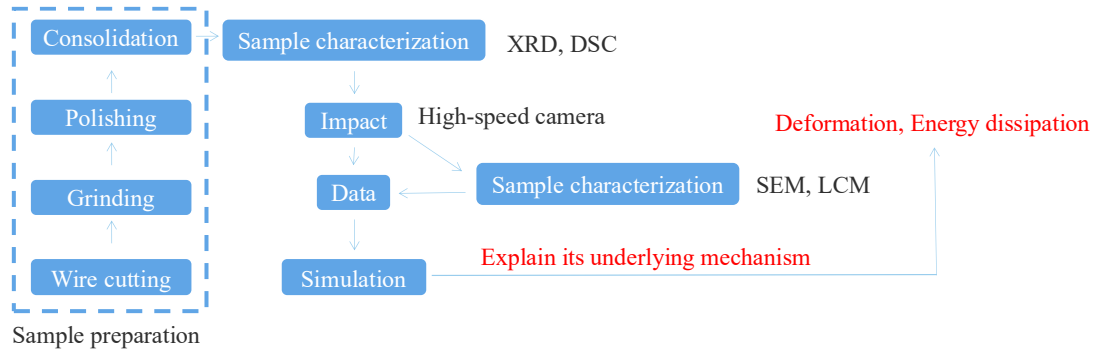


Figure 2 Experiment procedure

2.1 Sample Preparation and Characterization

Bulk samples of $\text{Pd}_{43}\text{Ni}_{10}\text{Cu}_{27}\text{P}_{20}$ (size: $10 \times 5 \times 3 \text{ mm}^3$) were selected as the targets. This Pd-based glass exhibits a low critical cooling rate, enabling the formation of bulk target. Also, the Gibbs free energy difference between its amorphous and crystalline states is only -0.1 kJ per gram-atom, indicating that it has a low driving for crystallization^[7]. Thus, it has a high amorphous structural stability, making it a preferable BMG for this study.

The BMG was cut to the required size using a diamond wire saw. After that, it is grinded and polished to achieve a smooth surface. Each sample was then affixed onto a glass plate which was connected to the temperature controller and consolidate. Steps of sample preparation are shown in Figure 3.

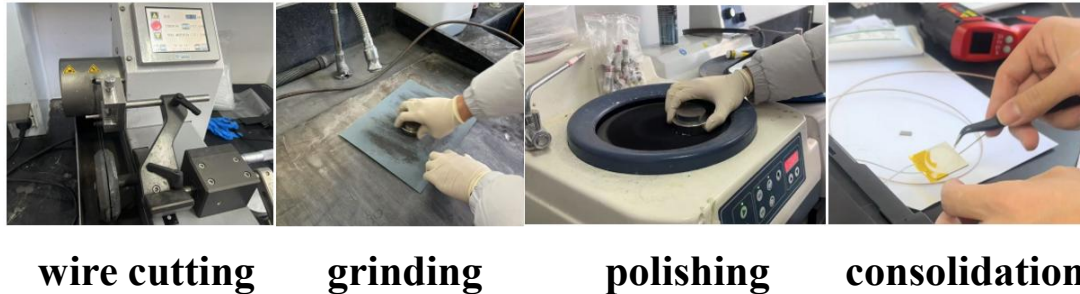


Figure 3 Steps of sample preparation

After sample preparation, the structure of the sample was characterized by X-ray Diffraction (XRD). The XRD instrument used in the experiment is SMARTLAB (9) (as shown in Figure 4a), it generates X-ray beam to illuminate the sample and the detector measures the angles and intensities of diffracted X-rays. The crystalline material exhibits sharp, distinct peaks while amorphous material presents broad, diffuse humps or only a general increase in background intensity. As shown in Figure 4b, the sample shows only one broad, diffused hump at around 40° , which is a characteristic amorphous scattering peak resulted in the cumulative scattering of all short-range atomic pairs. No sharp, narrow peaks are observed in the pattern, indicating that the atoms are only ordered over short distances and is lack of any long-range lattice. The inset to Fig. 4b shows the observation results of the transmission electron microscope (TEM), further indicating the disordered structure at the atomic scale. Hence, the sample is confirmed to remain in the glassy state and can be used in subsequent impact experiments.

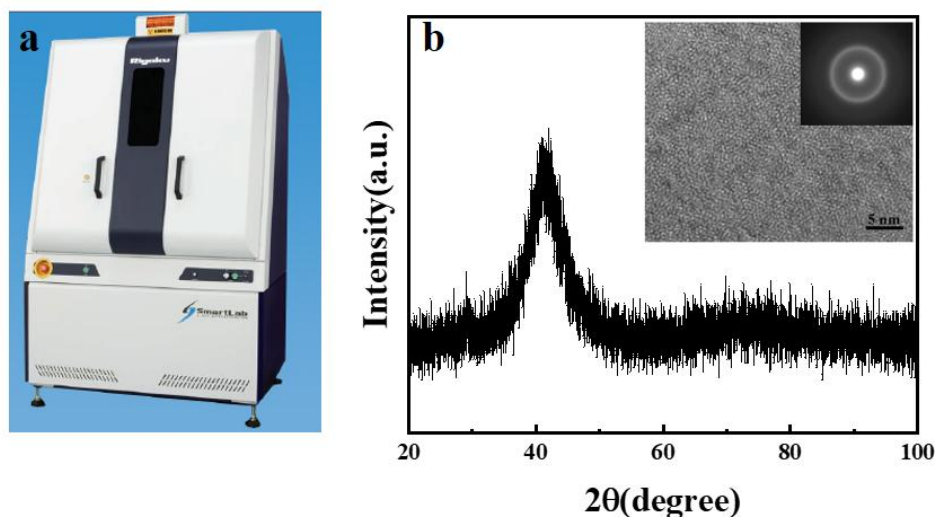


Figure 4 Structure characterizations of sample: **a** XRD experiment setup, **b** XRD pattern with the TEM observation

After structural characterization, samples then were characterized by Differential Scanning Calorimeter (DSC). The glass transition temperature (T_g) and crystallization temperature (T_x) were examined to make sure that experimental temperature does not exceed its crystallization temperature. The instrument used in characterization is STA449 F3 (see Figure 5a). It measures the difference in the heat flow into or out of a sample and reference, as a function of temperature. Before loading the sample, a blank scan was performed to calibrate baseline, thus the background noise is removed. The sample was then placed in the DSC sample pan and heated from 300 K to 750 K at a constant rate of 20 K/min. The heat flow versus temperature curve was plotted, as shown in Figure 5b.

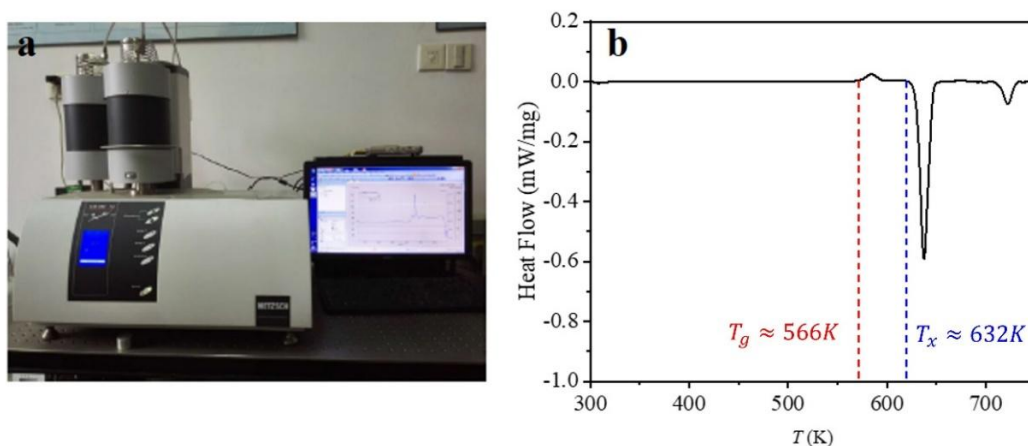


Figure 5 Differential scanning calorimetry characterization of target: **a** DSC experiment setup **b** DSC results indicating T_g and T_x

From the DSC curve, a baseline shift is observed at approximately 570 K (marked by the red dashed line), indicating the onset of the glass transition. During glass transition, the sample's heat capacity increases, so additional energy is required to maintain the set temperature increasing rate. As a result, an upward shift in the baseline of heat-flow signal is observed. The glass transition temperature of the sample was determined to be 570 K. Furthermore, a significant exothermic peak emerges at 632 K (marked by the blue dashed line), signifying the onset of crystallization. This exothermic peak results from the atomic rearrangement from chaotic to ordered, which lead to a negative enthalpy change. Based on these DSC data, the temperatures selected for the experimental groups (25 °C and 200 °C) are below the crystallization onset, ensuring that the sample retained its glassy structure throughout the tests.

2.2 Equipment Setup of Micro-particle Impact

LIPIT system is an equipment that is used to launch microparticles to impact target at various velocities. This technique is first proposed by Lee et al^[19]. Through adjusting laser parameters and the components of launching stage, impact speed ranged from 1~1000 m/s can be achieved. The LIPIT system used in this study is presented in Figure 6. This system consists of a laser emitter, a launching stage, a target, a high-speed camera, and an illumination source. The pulsed laser (wavelength 532 nm, duration 7 ns) is used. Between the laser and the launching stage, a convex lens is placed to focus the laser. The launching stage comprises a glass plate, an ablation (metallic) layer, and an elastic layer. The glass plate is made of K9 glass (thickness 4 mm), chosen for its high optical transmittance and low dispersion. It serves as the supporting carrier for both the ablation (metallic) layer and the elastic layer, and also provides physical constraints during ablation. The ablation layer consists of a 100 nm thick gold film, and the elastic layer is a polymer of 80 μm thick, to which the SiO_2 microparticles (30 μm in diameter) are attached. The target is mounted on a glass connected to a temperature-controller. On one side of the target a high-speed camera is positioned, and on the opposite side a laser light source is positioned. Together they capture the impact progress. The ultrahigh-frame-rate camera with the laser light source provides

nanosecond scale temporal resolution.

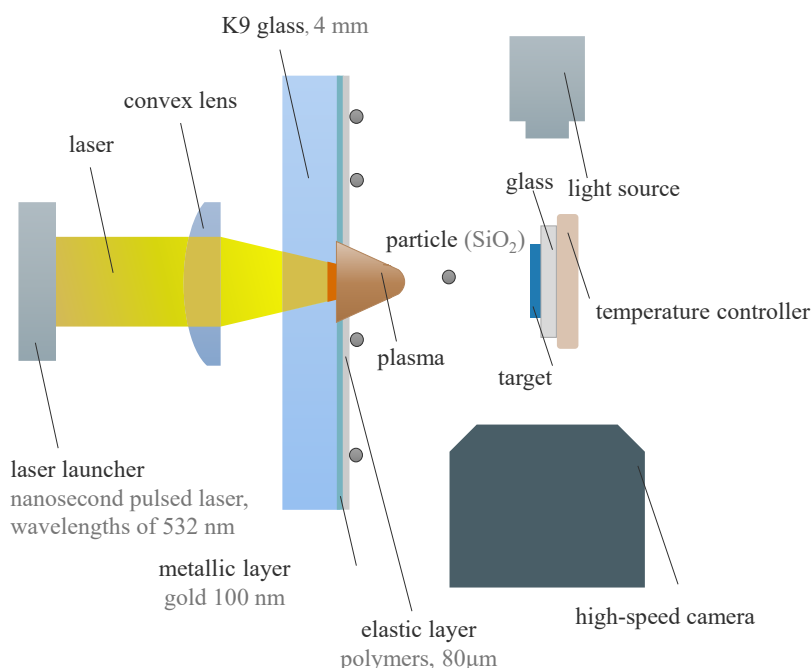


Figure 6 Schematics of the laser-induced micro-particle impact test platform

The temperature control device employed in this study is the REX-C900 intelligent PID temperature controller (shown in Figure 7a), which offers a wide temperature control range and high precision regulation. It is capable of controlling temperatures from 25°C to 400 °C, fully meeting the requirements of the experiment. The controller utilizes Proportional–Integral–Derivative (PID) technology to achieve accurate thermal regulation. Experimental results indicate that the temperature control accuracy can be maintained within ± 1 K, as illustrated in Figure 7b.

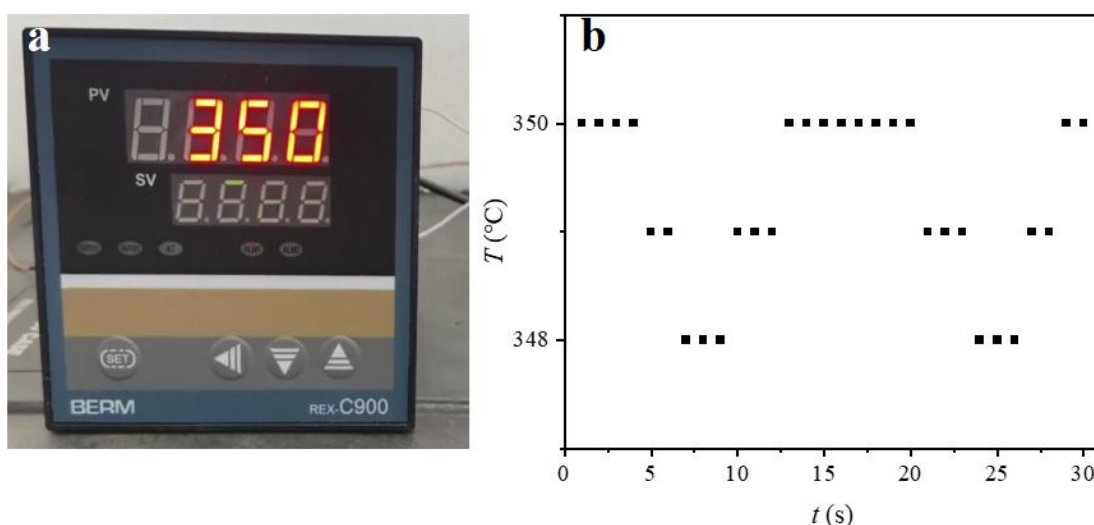


Figure 7 Temperature control: **a** device **b** accuracy

After the impact, SEM is used to observing surface morphology of the sample. The SEM was carried out by using Oxford X-Max N 80T (as shown in Figure 8a). The depth of the crater is characterized by laser confocal microscope (LCM). The LCM was carried out by using Olympus OLS4000 (see Figure 8b).

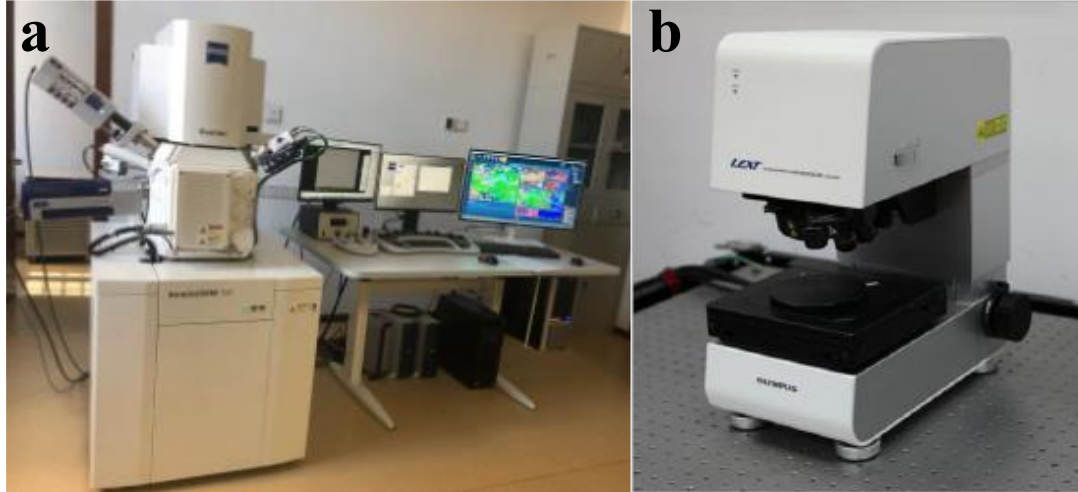


Figure 8 Scanning electron microscope (a) and laser confocal microscope setup (b)

2.3 Results and Discussion

2.3.1 In-situ Observation

By adjusting the laser pulse energy, particles were accelerated to different impact velocities. Tests were conducted at 25 °C, 100 °C and 200 °C conditions, with launch speeds spanning 70 m/s to 700 m/s. During every trial, a high-speed camera captured the particle's trajectory and its interaction with the target for subsequent analysis. From the in-situ observations (see Figure 9), it can be seen that under all temperatures, the post-impact particle displacements are substantially smaller than the pre-impact displacements. Thus, the rebound velocity of the particles is notably lower than their impact velocity, showing the system undergoes inelastic collisions. Furthermore, no particle adhesion was observed in either temperature group, the particle remains a ball shape after impact.

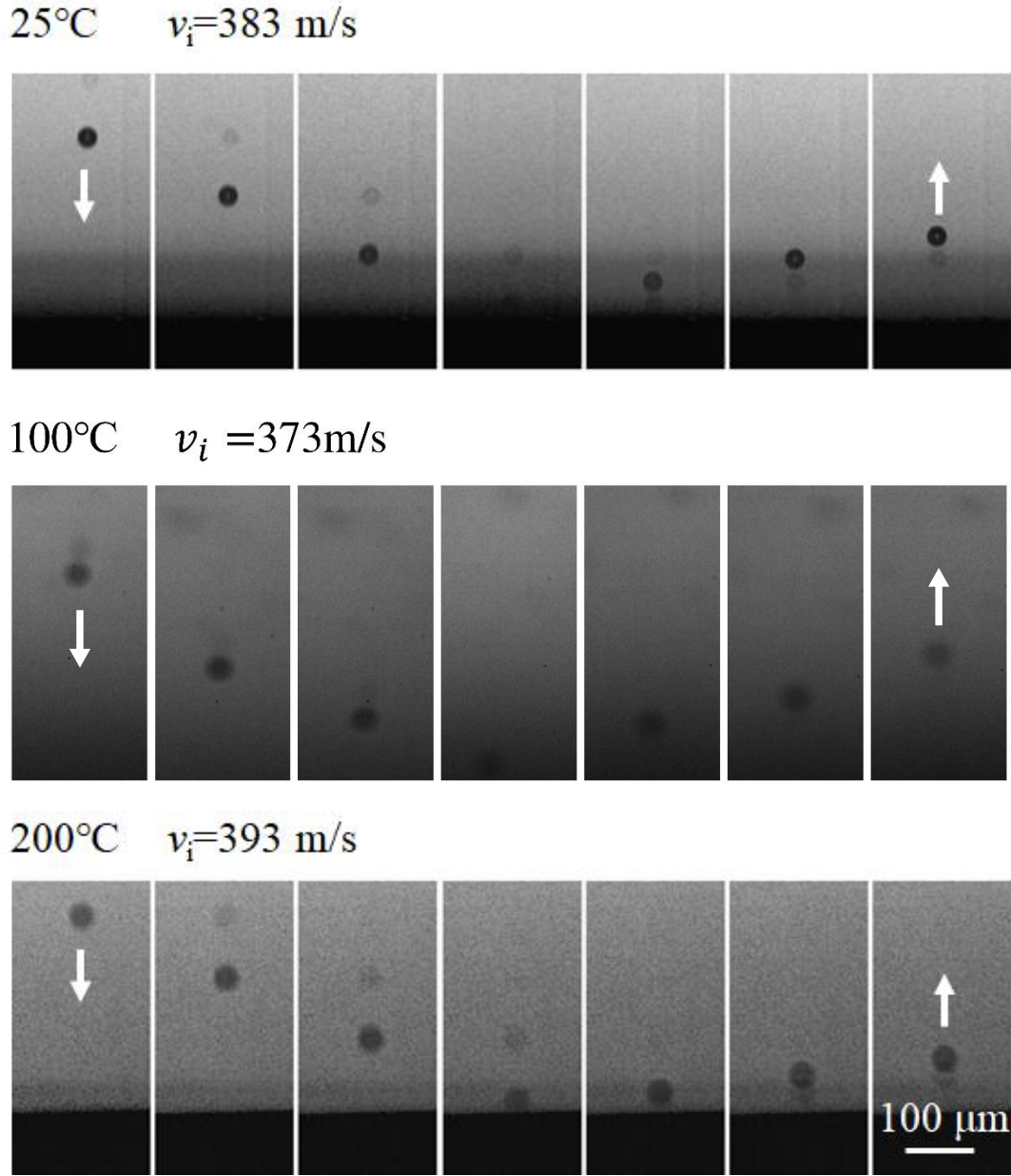


Figure 9 In situ observation at different temperature and velocities

From each test, the particle's impact velocity v_i and rebound velocity v_r were used to calculate the coefficient of restitution COR, defined by:

$$\text{COR} = \frac{v_r}{v_i} \quad (1)$$

The coefficient of restitution reflects energy dissipation of the target during the collision and is closely related to its impact resistance. Its value lies between 0 and 1: the closer COR is to 1, the more elastic the collision is; the closer to 0, the more inelastic it is. As impact velocity increases, COR tends to decrease, indicating greater kinetic energy loss through plastic deformation and stress wave propagation. COR decreases

according to different power-law regimes: it remains 1 during elastic collisions; then became proportional to $v_i^{-0.25}$ during the elastic-plastic deformation regime; then it became proportional to $v_i^{-0.5}$ during fully plastic deformation; it is eventually proportional to v_i^{-m} after transition into deep plastic deformation, m is a exponent depends on the material and usually greater than 0.5.

Figure 10 shows the coefficient of restitution for micro-particles impact on targets under different temperature. The COR at 200 °C are consistently lower than that at 25 °C. This indicates that at high temperature the material dissipates more energy. Therefore, the glass exhibits a better energy dissipation capability at high temperature. Conversely, at room temperature of 25°C, the higher COR values show that it undergoes less plastic deformation, indicating it has a better anti-penetration resistance. Moreover, in both temperature conditions, COR decreases as impact velocity increases, and it can be divided into two sections. In the first section, as well as velocity in 80-250 m/s range, the fitted results show that $COR = 8.6v_i^{-0.5}$ for 25°C and $COR = 7.7v_i^{-0.5}$ for 200°C, indicating that they both undergo finite plastic deformation. In the second section, velocity in the range of 250 m/s-700 m/s, COR decreases with a much steeper slope for targets under both temperatures. At 25°C, the fitted results in Figure 10 show that $COR = 140v_i^{-1}$. Compared to that at 25°C, COR of micro-particle impacts at 200°C decreases more sharply, that is, $COR = 372v_i^{-1.2}$, indicating that both targets transformation to the deep plastic deformation regime.

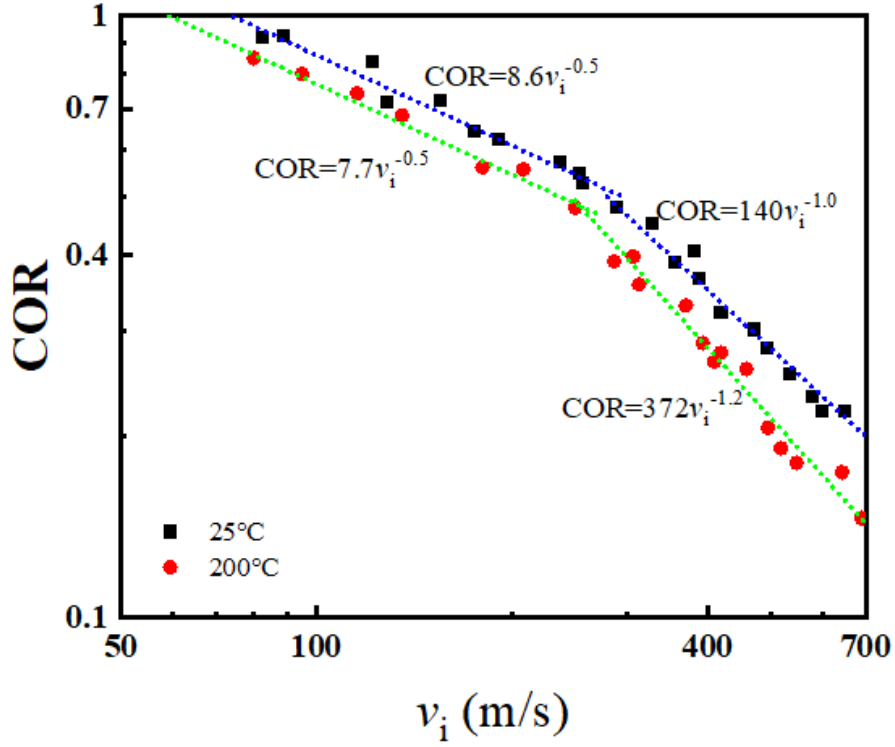


Figure 10 Coefficient of restitution for micro-particles impact on glassy targets under different temperatures.

2.3.2 Post Impact Observation

The impact crater features were further observed by SEM to uncover the dissipation modes of the impact energy. Figure 11 show the typical crater morphologies of targets at 25°C and 200°C, respectively. The circular craters were observed in both targets. Relatively smooth 'flat area' lies in the middle of the crater. However, several annular bands are observed on the inner wall of craters at both temperatures. The origin of these patterns is the initiation and development of shear bands in amorphous targets, which is a significant part in the energy dissipation mechanism of metallic glasses. There is little difference in the deformation between two temperatures. Besides uniform plastic deformation and shear bands, both targets have regular crater shape, with smooth edges and no pile-up. This implies that both samples at relatively low impact velocity still maintain a solid phase.

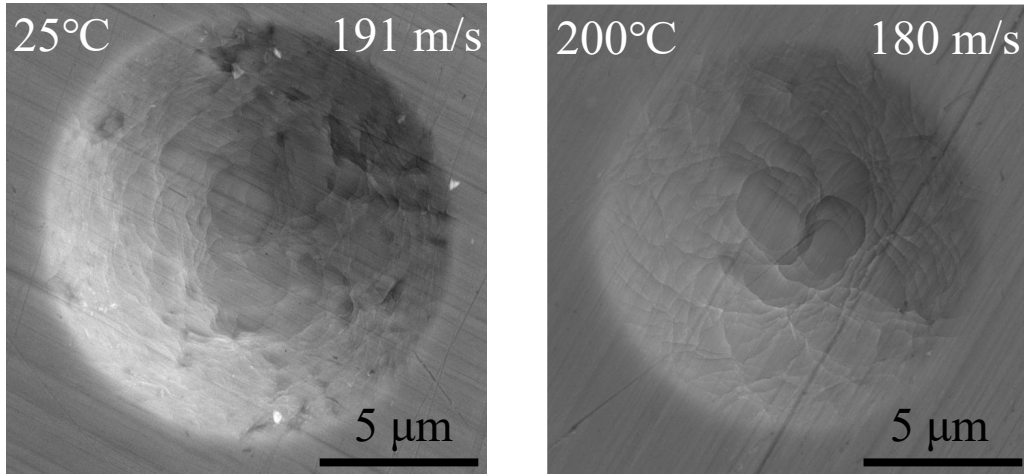


Figure 11 Postmortem observation of the impact craters at low impact velocities.

Figure 12 shows two typical surface morphologies at relatively high impact velocity. This indicates that at higher temperature the BMG exhibit a larger deformation than that at lower temperature, which leads to a more remarkable energy dissipation and a lower COR value. In addition, both samples exhibit pile-up at the edge of the crater, while this phenomenon is more obvious and notable at higher temperature. The pile-up material is higher and its volume is larger in higher temperature. The pile up is slightly lower at low temperature, and the shape of it is regular, while the shape of pile up at high temperature is highly irregular, resembling an outward-splash pattern. This shows that the BMG, at high velocity impact, exhibits the characteristics of highly viscous fluid at high temperature of 200°C, while maintains solid characteristics at low temperature of 25°C. This phenomenon might be caused by the adiabatic temperature rise. Because of the adiabatic temperature rise, the instantaneous temperature of glassy samples might be close to or even exceed T_g . Due to the unique structural dynamics, they exhibit the characteristics of viscous flowing supercooled liquid region, which explain the pile-up pattern in Figure 12.

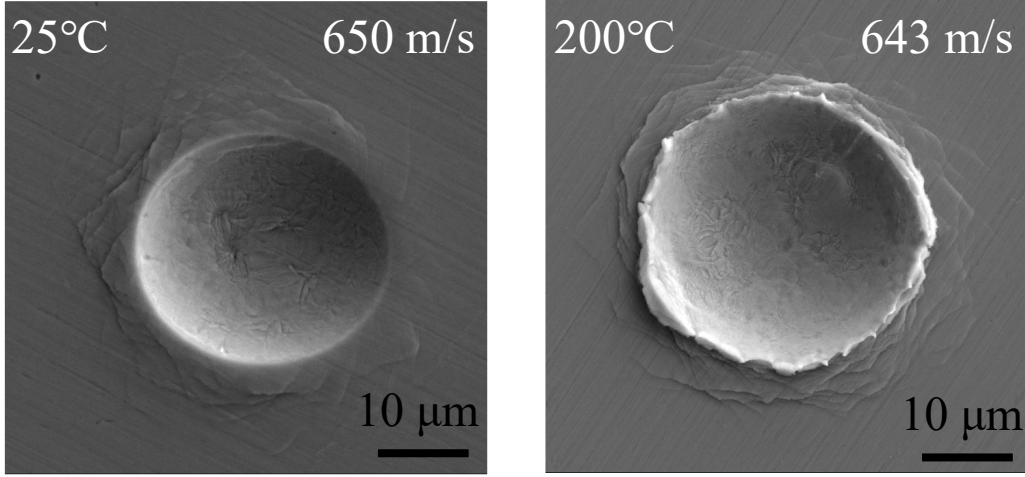


Figure 12 Postmortem observation of the impact craters at high impact velocities.

COR also reflects the mechanical property difference of materials under different temperatures. Based on the COR data, both groups of targets undergo finite plastic deformation regime, exhibiting COR proportional to $v_i^{-0.5}$. When the samples are in finite plastic regime, according to FEM results by Wu et al.^[7], COR satisfies the following equation:

$$\text{COR} = \alpha \left(\frac{v_i/v_y}{E^*/Y_d} \right)^{-\frac{1}{2}} \quad (2)$$

where v_i is the impact velocity; α is an empirical coefficient, taken 0.78 in this study; v_y is the yield velocity, i.e., at which plastic deformation is just initiated, and is given by Johnson and Li et al.^[9]:

$$v_y = 5.052 \left(\frac{Y_d^5}{E^{*4} \rho} \right)^{\frac{1}{2}} \quad (3)$$

where ρ is the particle density; Y in both equations represents the dynamic yield strength; E^* is the representative Young's Modulus and is given by:

$$\frac{1}{E^*} = \frac{1 - \mu_p^2}{E_p} + \frac{1 - \mu_t^2}{E_t} \quad (4)$$

where E and μ represent the Young's modulus and Poisson ratio of particle and target, respectively. By substituting equation (3) into (2), the formula for dynamic yield strength Y_d can be solved as:

$$Y_d = \left(\frac{\rho E_*^2 v_r^4}{26 v_i^2 \alpha^4} \right)^{\frac{1}{3}} \quad (5)$$

By using equation (5), the dynamic yield strength of BMGs at 200°C, 100 °C and 25°C in finite plastic deformation regime can be calculated. The results are shown in Figure 13. The average dynamic yield strength at 25°C is 2089 ± 110 MPa, while it is approximately 1901 MPa at 100 °C and 1825 ± 65 MPa at 200°C, showing that the average dynamic yield strength in finite plastic deformation regime decreases when temperature increases. Specifically, it decreases by 10.39% from 25°C to 200°C. Such significant decrease in dynamic yield strength discloses that BMGs at high temperature perform a larger plastic deformation to dissipate energy, which conforms the previous analysis that the BMG exhibits a better energy dissipation mechanism under high temperature while has a better impact resistance under low temperature.

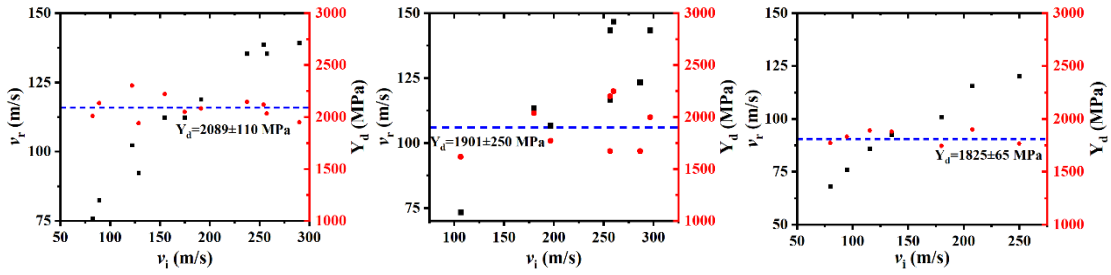


Figure 13 Dynamic yield strength of targets under different temperatures

3. Simulation

3.1 Constitutive Model

After conducting the experiment, numerical simulations were conducted to analyze the underlying mechanism of BMGs' impact response. The experimental results show that the plastic deformation dissipates significant energy. In order to further reveal the potential mechanism of macroscopic physical phenomena, the FEM simulation was carried out by using the coupled shear transformation zones (STZs)-free volume constitutive model.

Over the past decades, the constitutive model of plastic flow of metallic glasses can be clarified into two categories: “free volume” model^[10~12], which contends that

monoatomic jump is the basic movement under strain motivation; “shear transformation” model^[13~15], which considers plastic flow as the result of shear movements cooperated by atomic clusters composed of tens to hundreds of atoms. However, further research showed that plastic flow relates to both free volume and shear transformations^{[16][17]}, since shear transformation is more likely to occur in regions with sufficient free volume; consequently, shear transformation activates local free volume. Based on this mechanism, Jiang et al.^[18] considered the coupling effects of free volume and shear transformation to develop a STZ-free volume coupling model for metallic glasses.

In order to extend the model to multiaxial stress conditions, equivalent stress is introduced here. Therefore, the constitutive model of BMGs under three-dimensional stress state is described as follows. All variables used in the equations are nondimensional:

$$\dot{\varepsilon}_{ij}^{pl} = \exp\left(-\frac{1}{\xi}\right) \exp\left[E_A\left(1 - \frac{1}{T}\right)\right] \left[\Lambda \sinh\left(\frac{\sigma'_e}{T}\right) - \Delta \cosh\left(\frac{\sigma'_e}{T}\right)\right] \frac{S_{ij}}{\sigma'_e} \quad (6)$$

$$\dot{\sigma}_{ij} = \varepsilon_0 \mu (\dot{\varepsilon}_{ij} - \dot{\varepsilon}_{ij}^{pl}) \quad (7)$$

$$\dot{\Delta} = 2\dot{\varepsilon}_e^{pl} - \phi |\sigma'_e \dot{\varepsilon}_e^{pl}| \Delta \quad (8)$$

$$\dot{\Lambda} = \phi |\sigma'_e \dot{\varepsilon}_e^{pl}| (1 - \Lambda) \quad (9)$$

$$\dot{\xi} = D_{ia} |\sigma'_e \dot{\varepsilon}_e^{pl}| - \varphi \exp\left(-\frac{1}{\xi}\right) \exp\left[E_a\left(1 - \frac{1}{T}\right)\right] \quad (10)$$

where ε is strain and $\dot{\varepsilon}$ is strain rate, which can be divided into elastic part $\dot{\varepsilon}_{ij}^{el}$ and plastic part $\dot{\varepsilon}_{ij}^{pl}$. Elastic deformation strictly obeys Hook's Law, $\sigma = \mu \varepsilon^{el}$; Plastic deformation is contributed by a series of shear transformation. ξ represents free volume, T represents temperature, $E_A = \Delta G/k_B T_g$ is the activation energy of thermal activation. σ'_e is the equivalent stress, given by $\sigma'_e = \Gamma \sigma_m + \sqrt{J_2}$, with hydrostatic pressure sensitivity factor Γ and the Cauchy stress deviatoric stress tensor $S_{ij} = \sigma_{ij} - \delta_{ij} \sigma_m$. The shear transformations have two statuses: before transformation and after transformation, and they alter between these two statuses. Δ

and Λ are normalized difference and sum of shear transformation number density of two statuses, ϕ is a dimensionless coefficient. The evolution of free volume is the consequence of dynamic competition between the shear transformation motivation and relaxation-induced annihilation. Therefore, the evolution of free volume depends on both thermal-activated shear transformation and free volume annihilation rate, as represented in equation (10). D_{ia} in the equation is dilatancy factor for measuring system's capacity of generating free volume, ϕ is coefficient measuring free volume annihilation capacity, E_a is activation energy of free volume thermal annihilation.

FEM simulations were conducted using ABAQUS commercial software. Due to the symmetry, only a quarter-symmetrical model is necessary (see Figure 14). The radius of the micro-particle is 0.015 mm. The size of the bulk target is $2 \times 2 \times 1 \text{ mm}^3$, respectively. Surface-to-surface contact is used between the micro-particle and the target. The detailed material parameters used is listed in Table 1. The C3D8I element was adopted and the number of the elements is about 1.35 million. The minimum element size at the contact region being $0.25 \times 0.25 \times 0.25 \text{ }\mu\text{m}^3$.

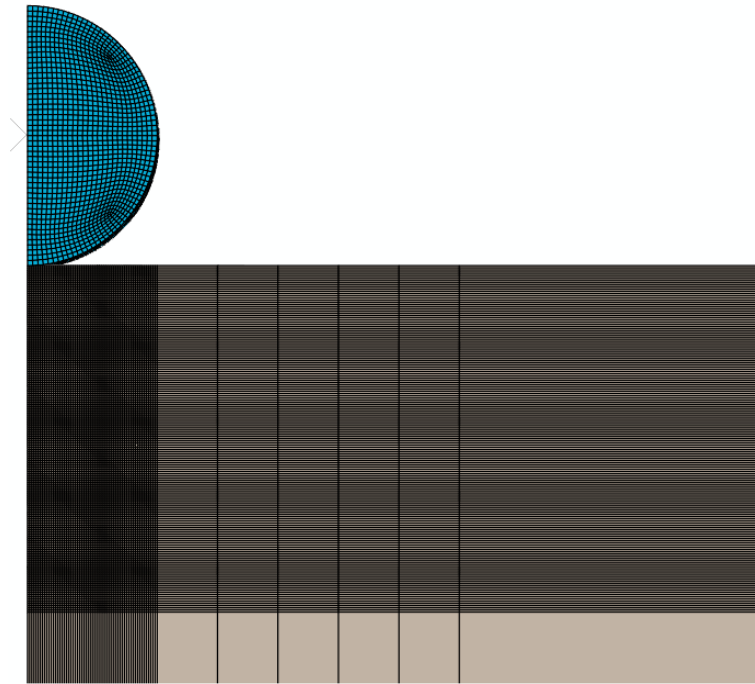


Figure 14 The FEM model for micro-particle impacting on the BMG.

Table 1 material parameters used in simulations

Parameters	Value
Young's modulus of BMG	96 GPa
Density of BMG	6125 Kg·m ⁻³
Young's modulus of micro-particle	63 GPa
Density of micro-particle	2200 Kg·m ⁻³
Poisson ratio of BMG and micro-particle	0.3
Initial free volume concentration	0.05
Initial bias of normalized population of STZ	0
Initial summation of normalized population of STZ	0.5
Glass transition temperature	625 K
Reference STZ activation energy	$G_0(T)$
STZ activation volume	0.1 nm ³
Steady-state population of STZ	10 nm ⁻³
Ratio of the coefficient $\bar{\phi}$ and the geometrical factor χ	10
Proportional coefficient between STZ-creation/annihilation rate and plastic work rate	0.5
Dilatancy factor	0.01
Attempt frequency	10 ¹³ s ⁻¹
Boltzmann's constant	1.38×10 ⁻²³ J·K ⁻¹

3.2 Results and Discussion

Figure 15 shows the deformation of the target at 200 °C when $v_i=300$ m/s. The red solid line is the result of LCM and the blue solid line is the result of FEM. It can be seen that the results of FEM are in good agreement with the experimental results. Therefore, the present model can be used for further analysis.

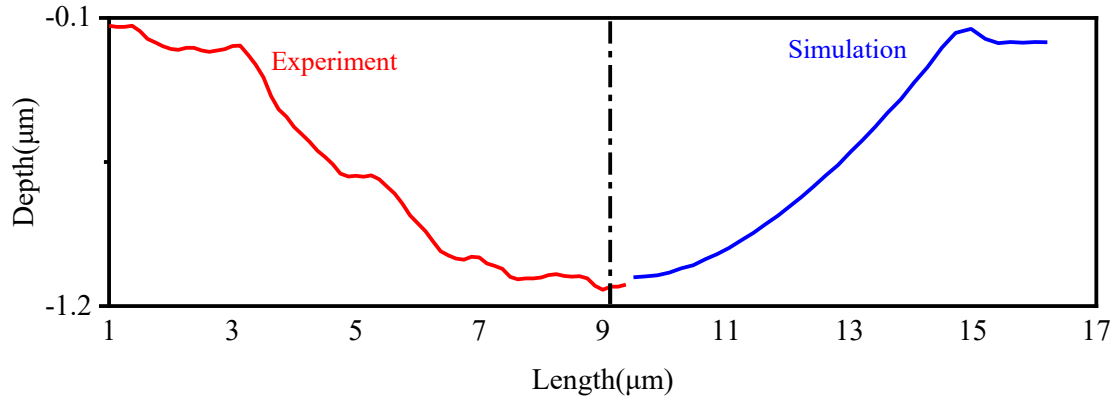


Figure 15 Comparison between experiment and simulation: the displacement of the micro-particle impacting on the target.

In order to further analyze the difference between the two processes, the impact of different temperatures at $v_i=400$ m/s was simulated. The displacement cloud images of the characteristic time points are extracted respectively, as shown in Figure 16. The second and fourth columns correspond to the moments of the deepest indentation and the ball-target detachment, respectively. At the initial loading ($t = 8$ ns), there was no significant difference between the two targets. However, at 25 °C, when $t = 14.4$ ns, the indentation is the deepest, and it takes 15.2 ns at 200 °C. In addition, the ball-target detachment time is 23.6 ns at 25 °C and 25.6 ns at 200 °C. This indicates that the projectile has a longer interaction time with the target at higher temperatures, resulting in longer energy transfer. At the same time, it can be seen from the deformation cloud image that the cloud image color of the crater center at 200 °C is significantly deeper when the indentation is the deepest, which indicates that it has greater deformation. After the impact, the deformation zone of the spherical region is formed inside the two targets.

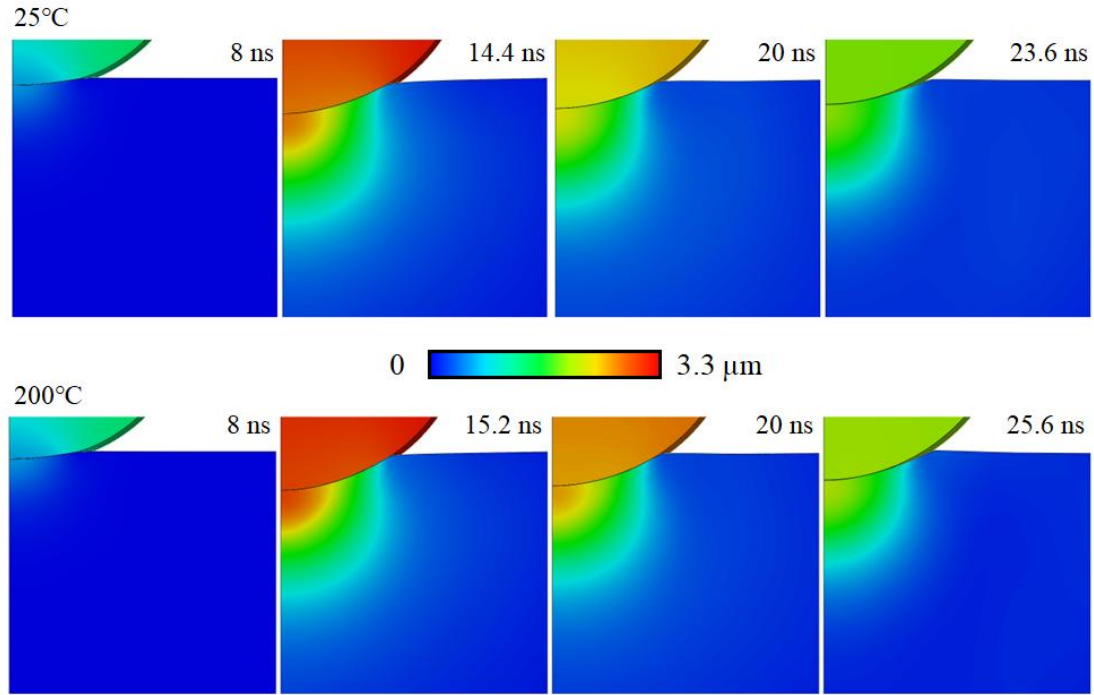


Figure 16 The indentation progress under different temperature at characteristic times.

Figure 17 further gives the field average results of different physical quantities. Here, the region used for averaging is selected as the region of plastic deformation, as the elastic region with too small deformation will recover after the projectile is separated from the target, so it cannot be used to compare the deformation and mechanism differences between the two targets. It can be seen from the field average results of stress that the stress at 200 °C is significantly smaller than that at 20 °C, which further confirms the softening phenomenon of the target observed in the experiment. The results of equivalent plastic strain show that there is little difference between the two targets during the initial loading stage. However, with the increase of the indentation depth, the deformation growth at 25 °C slows down and quickly reaches equilibrium, while the deformation growth at 200 °C exhibits two stages: first rapid growth and then slow growth. This indicates that the plastic flow of the material is more significant at higher temperatures. The normalized STZ number density shows that the sum of the STZ number density in the plastic zone of the two targets almost shows the same change trend, but the difference shows a completely different change trend. The difference in the number density of STZs at 200 °C is only significantly higher, indicating that more positive shear transformation occurs. It is also noted that the

variation trend of this quantity is the same as that of the equivalent plastic strain. Therefore, high temperature promotes the shear transformations, resulting in greater plastic strain.

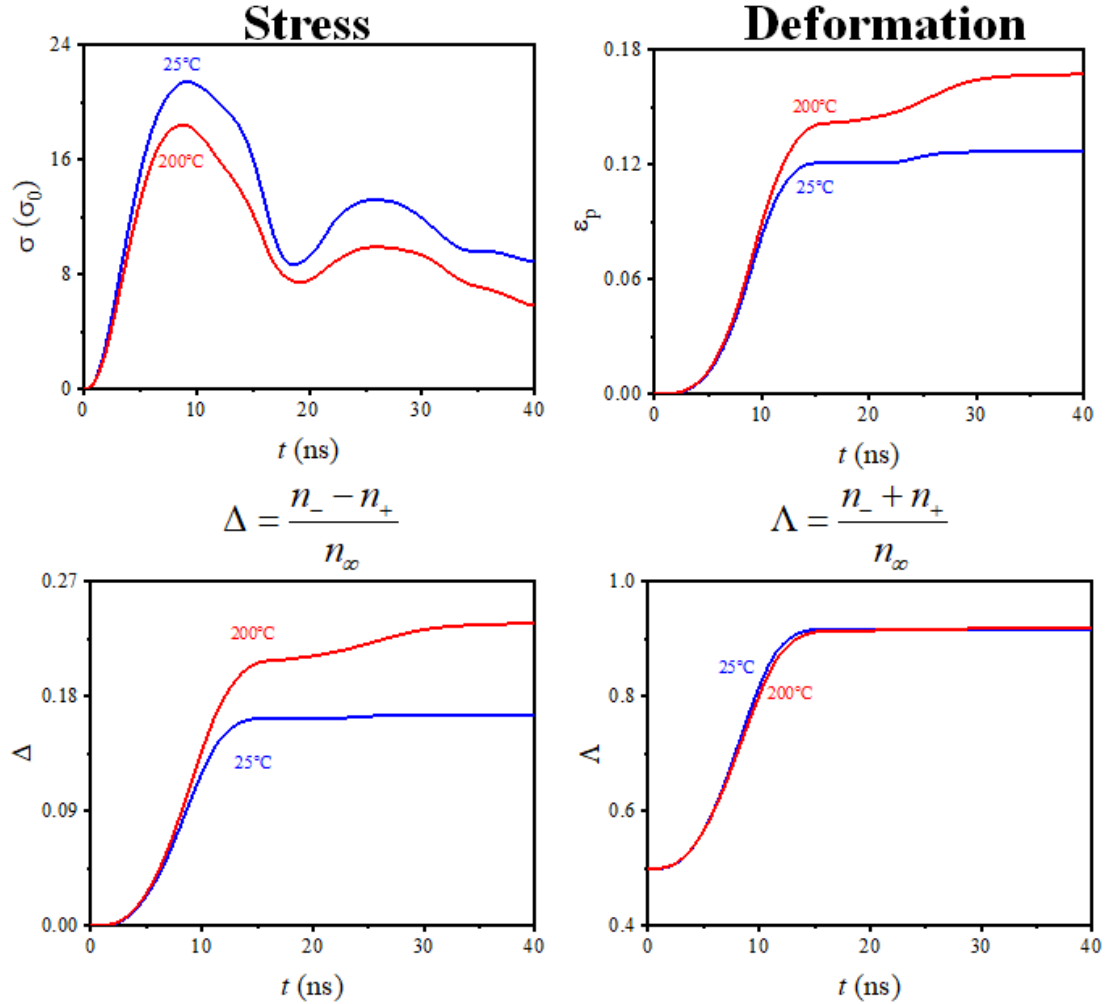


Figure 17 Field-averaged results of different physical quantities under different temperatures.

4. Limitations

Although the study demonstrates several interesting results, there are some limitations in the design and conduction of the study.

First, the dynamic yield stress is not measured directly but inferred from particle velocity and crater geometry. This involves several measured quantities with finite precision, such as particle size, velocity, which may involve uncertainties. Small errors

in velocity would produce a larger relative error in kinetic energy ($\propto v^2$) and thus in the calculated stress.

Second, during each experiment, although the input temperatures (20°C, 100°C, 200°C) stay the same by the control of PID temperature control system, thermal gradients (through the sample thickness or across its surface) may exist. Any unmodeled heat transfer or cooling of the stage could shift the actual impact temperature. However, because of the limits of equipment, the in-situ temperature is hard to be monitored at the impact site. Thus, the effective temperature during impact has uncertainty, which would affect material properties and flow stress.

Third, since the in-situ temperature cannot be measured in experiments, the adiabatic heating during impact cannot be precisely measured, and this phenomenon is not the main behavior this study wants to investigate. Thus, the study's simulation primarily focuses on the change in its shear transformation but didn't account for the influence of the transient thermal spike. The work done by LH Dai et al. had revealed the effects of adiabatic shear banding in BMGs, providing solid support for the adiabatic heating during impact.

Fourth, during heating process, the metallic surface may easily react with the air and being oxidized. Such oxides alter surface strength, adhesion, and flow. By experiment, it was found that heating BMGs to temperatures above its glass transition temperature will cause severe oxidation, making it unsuitable for the investigation. So, the temperature range was limited to 200°C highest. Although this temperature range ensures no severe oxidation, as verified by element testing, it limits the range of the investigation.

5. Conclusion

This study performed the experimental investigations and FEM simulations on the high-velocity microparticle impact response of Pd₄₃Ni₁₀Cu₂₇P₂₀ BMG at 25°C, 100°C and 200°C. We obtained three key conclusions as follows:

First, the COR decreases with increasing impact velocity under both temperatures,

indicating enhanced energy dissipation at higher velocities. Post-impact morphological analysis via SEM reveals that at high impact velocities, the BMG at 200°C shows larger craters with more pronounced pile-up resembling fluid-like splash patterns, suggesting a transition toward viscous flow behavior due to adiabatic heating near T_g .

Second, the dynamic yield strength of the BMG decreases when temperature increases, specifically by approximately 10.39% when the temperature increases from 25°C to 200°C, confirming the material's thermal softening effect and consistent with the enhanced plastic deformation and energy dissipation observed at high temperature.

Third, FEM simulations using a physical model successfully captures the impact response and crater profiles observed experimentally. The simulations demonstrate longer interaction times, greater plastic strain, and easier shear transformations activation at 200°C.

This study underscores the significant influence of temperature on the impact behavior of metallic glasses, highlighting their transition from solid-like to fluid-like response under high-strain-rate loading. The findings contribute to a deeper understanding of the dynamic mechanical properties of amorphous materials and offer valuable insights for the design of high-performance BMG components under extreme conditions.

6. Reference

- [1] Scholze, H. (2012). Glass: nature, structure, and properties, P45-48. *Springer Science & Business Media*.
- [2] Wang, W. H. (2011). A Brief History of Metal Glass Research. *Physics*, 40(11), 701-709.
- [3] Guo, Y. C., Wang, Z. X. (1984). Amorphous State Physics, *Beijing: Science Press*
- [4] Wang, W. H., Dong, C., Shek, C. H. (2004). Bulk metallic glasses. *Materials Science and Engineering: R: Reports*, 44(2-3), 45-89.
- [5] Dowding, I., Schuh, C. A. (2024). Metals strengthen with increasing temperature at extreme strain rates. *Nature*, 630(8015), 91-95.
- [6] Sakino, K. (2000). Transition in rate controlling mechanism of FFC metals at very high strain rates and high temperatures. *Le Journal de Physique IV*, 10(PR9), Pr9-57.
- [7] Kuno, M. (2001). Thermodynamics of the Pd₄₃Ni₁₀Cu₂₇P₂₀ metallic glass-forming alloy. Oregon State University.
- [8] Wu, C. Y., Li, L. Y., Thornton, C. (2003). Rebound behaviour of spheres for plastic impacts. *International journal of impact engineering*, 28(9), 929-946.
- [9] Johnson, K. L. (1987). Contact mechanics. Cambridge university press.
- [10] Spaepen, F. (1977). A microscopic mechanism for steady state inhomogeneous flow in metallic glasses. *Acta metallurgica*, 25(4), 407-415.
- [11] Jiang, M. Q., Dai, L. (2007). Intrinsic correlation between fragility and bulk modulus in metallic glasses. *Physical Review B—Condensed Matter and Materials Physics*, 76(5), 054204.
- [12] Jiang, M.Q., Jiang, S.Y., Dai, L.H. (2009). Inherent shear-dilatation coexistence in metallic glass. *Chinese Physics Letters*, 26(1), 016103.
- [13] Argon, A. S. (1979). Plastic deformation in metallic glasses. *Acta metallurgica*, 27(1), 47-58.
- [14] Şopu, D., Stukowski, A., Stoica, M., Scudino, S. (2017). Atomic-level processes of shear band nucleation in metallic glasses. *Physical review letters*, 119(19), 195503.
- [15] Xu, B., Falk, M. L., Li, J. F., & Kong, L. T. (2018). Predicting shear transformation events in metallic glasses. *Physical review letters*, 120(12), 125503.
- [16] Langer, J. S. (2001). Microstructural shear localization in plastic deformation of amorphous solids. *Physical Review E*, 64(1), 011504.

- [17] Lemaître, A. (2002). Rearrangements and dilatancy for sheared dense materials. *Physical Review Letters*, 89(19), 195503.
- [18] Jiang, M. Q., Wilde, G., & Dai, L. H. (2015). Origin of stress overshoot in amorphous solids. *Mechanics of Materials*, 81, 72-83.
- [19] Lee, J. H., Veyssset, D., Singer, J. P., Retsch, M., Saini, G., Pezeril, T., Thomas, E. L. (2012). High strain rate deformation of layered nanocomposites. *Nature communications*, 3(1), 1164.
- [20] Jdrewitt. (2021). Image: Silica. Wikipedia.
- [21] Fukushima, T., Shinya, H., Fujii, H., Sato, K., Katayama-Yoshida, H., & Dederichs, P. (2014). First-principles studies of GeTe-based dilute magnetic semiconductors. *Journal of Physics: Condensed Matter*, 27(1), 015501.

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When learning about "melting point" in middle school, the characteristic that glass doesn't have a specific melting point intrigues me. During Medieval, glass once was even considered "liquid". In high school, I learned that glass is an amorphous alloy that has a disordered molecular configuration at the microscale. Metals, on the other hand, are typical crystalloids, which have an ordered atom configuration. Driven by curiosity, I noticed the existence of an amorphous alloy, which is an alloy that has a disordered atomic configuration, like glass. Since glass and metal are distinct in atom configurations and macro properties, I wonder what mechanical properties the combination of these two materials will appear. When hit by particles, would an amorphous alloy shatter like glass does, or would it undergo plastic deformation like metal does? More importantly, is there any other variable, like temperature, that will affect its behaviors? Driven by this question, I first read lots of research paper to enrich my knowledge as well as studying fundamental knowledge of material science. Here, I again sincerely thank Professor Jiang and Teacher Jing for helping me when I was puzzled.

After preliminary research, the experiment was conducted. With the help of Professor Jiang, I learned to prepare samples and to operate the LIPIT system. At first, I decided to observe and analyze the amorphous alloy's behavior when impacted by particles at different temperatures. However, I discovered that when the sample temperature exceeds 550K, the alloy will be oxidized and form a thin membrane that affect the experiment. After discussing with the teacher, we modified

the temperature controller so that it can control temperature more precisely to make sure it doesn't exceed 550K. We once tried to carry out the experiment in a vacuum environment, but the vacuum space required was too large to be created, so we ultimately changed the approach. My role in experimenting is to prepare samples and conduct formal experiments. The supervising professor first tutored me to use the LIPIT system, then he and teacher Xiaohui ensured the security of the experiment by supervising me when I did experiments. Another big difficulty I encountered during the experiment was that the laser sometimes cannot accurately aim the particles, resulting in some particles remaining on the elastic layer rather than be launched towards the target. In order to solve that, I manipulated the laser launcher on the computer to make sure the laser path and the particle aligned as closely as possible.

During sample characterizations, I used the X-ray diffraction, differential scanning calorimeter, and scanning electron microscope with the tutorial of Professor Jiang. After gaining the primary data, I used Origin software to process them and obtained graphs.

As for data analysis, I used the high-speed camera to collect in-situ observation of the experiment, and used Kirana to analyze these results to obtain impact and rebound velocity for each trial. I used Origin Pro to graph the data and to draw graphs of the coefficient of restitution and dynamic yield strength to further analyze the mechanism of the sample. Professor Jiang lent me some invaluable feedback after I finished the primary analysis. I then revised and improved my analysis to make it more precise and comprehensive.

After experimentation and analysis, I want to further understand the underlying mechanism of the liquid-like behavior. Therefore, the simulation was conducted. I looked through a range of paper and found two main models for this behavior: "free volume" and "shear transformation". Thus, I decided to use a shear transformation zones-free volume coupling model that was proposed by Professor Jiang in 2015 to conduct finite element simulations.

As for essay writing, I independently finished the first draft. After that, I talked to my supervising professor Jiang about my opinions and confusions, we exchanged ideas, and these talks gave me further insights into the research, helping me to come to a more structured and solid essay.

Once again, I would like to extend my profound gratitude to all the people who have offered me invaluable help. It was their supports that made this journey both colorful and inspiring; It was their guidance that ignited my pathway to scientific exploration.